

DFMEA — Candidate E2

Failure mode	Effect	Cause	Current control	Severity	Occurrence	Detection	RPN	Mitigation / residual note
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Thermal overshoot	Cell temperature rise	High discharge current	Current limit	High	Medium	PyBaMM model	100	Design review
Lithium plating	Capacity fade	Aggressive fast charge	Charge protocol	High	Low	PyBaMM model	50	Candidate E2 maintains
Energy density risk	Failure to meet vehicle range	Overly conservative design	Design review	Medium	Medium	Contract-caliber energy calculation	100	Final design exceeds 50
Model-form risk	PyBaMM prediction error	Simplified model	Model validation	Medium	Medium	Automated model validation	100	Report and log preserved